Junseok Oh (오준석)

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Professional Experience

Samsung Fire & Marine Insurance Present
Pro, AI Development Part, AX/PI Center

Education

Sogang University Mar 2022 – Aug 2026
Seoul, South Korea

Ph.D. in Computer Science and Engineering (Advisor: Prof. Ji-Hwan Kim)

- Dissertation: "Adapter-Only Bridging of Frozen Speech Encoder and Frozen LLM for ASR"

Sogang University Sep 2017 – Aug 2019
Seoul, South Korea

Master of Engineering in Computer Science and Engineering (Advisor: Prof. Ji-Hwan Kim)

- Thesis: "Korean Real-Time Automatic Transcription System Using Weakly Labeled Corpus"

Sogang University Mar 2010 – Aug 2017
Seoul, South Korea

Bachelor of Engineering in Computer Science and Engineering

Uijeongbu High School Mar 2007 – Feb 2010
Uijeongbu, Gyeonggi-do, South Korea

High School Diploma

Research Experience

Development of Dialog-based Multi-modal Explainable AI Technology *Apr 2022 – Present*
Funding Agency: Ministry of Science and ICT (MSIT) / IITP

- Developed an AI-based framework for dysarthria (speech disorder) severity classification, providing multi-modal explanations to support diagnostic decision-making.
- Implemented speech-based explainable diagnostic modules that analyze acoustic and linguistic features to provide interpretable feedback on severity levels.
- Researched methods to bridge the gap between complex AI model outputs and user-understandable explanations in a multi-modal dialog environment.

End-to-End Korean Speech Recognition Project *2024 – Apr 2025*
Industry Partner: Smilegate

- Developed a universal Korean ASR system using a hybrid FastConformer RNN-Transducer (RNNT) + CTC model.
- Implemented a cache-aware streaming method to improve recognition accuracy and reduce latency.
- Implemented context biasing to adapt the ASR decoder for domain-specific vocabulary, such as gaming terminology, thereby improving recognition accuracy and robustness in specialized contexts.

End-to-End Speech Recognition for Telephony *Apr 2024 – Dec 2024*
Industry Partner: LOTTE INNOVATE COMPANY

- Developed streaming and non-streaming Korean ASR pipelines optimized for 8 kHz telephony data using a FastConformer-CTC architecture.
- Implemented context-biasing modules to dynamically adapt the decoder's vocabulary for domain shift, boosting word-level accuracy.

Automated Korean Speaking Assessment (2024) *May 2024 – Dec 2024*
Funding Agency: Ministry of Culture, Sports and Tourism of South Korea

- Developed a multi-task learning framework using Wav2Vec to jointly model pronunciation, fluency, and content for L2-Korean assessment.
- Integrated Conformer-CTC ASR outputs with a large language model (LLM) to generate automated, multi-aspect scoring.

Intelligent Analysis and Classification-based Content Rating Technology to Address Uncontrolled Distribution of Harmful Media *2022 – 2024*
Funding Agency: Ministry of Science and ICT of South Korea

- Led the audio analytics submodule within an automated video content rating framework, designing sound-event detection models for key multimedia events.
- Fine-tuned OpenAI's Whisper ASR model on domain-specific video corpora to enhance transcription robustness in diverse acoustic environments.

Automated Korean Speaking Assessment (2023) *May 2023 – Dec 2023*
Funding Agency: Ministry of Culture, Sports and Tourism of South Korea

- Built an end-to-end evaluation pipeline for L2 Korean speakers by combining Conformer-CTC ASR outputs with BERT-based semantic scoring.
- Developed algorithms to quantify pronunciation accuracy, speech rate, and syntactic correctness from ASR transcripts.

Development of a Video Story Understanding-based QA System to Pass the Video Turing Test *Sep 2017 – Dec 2019*
Funding Agency: Ministry of Science and ICT of South Korea

- Modified Kaldi's sentence-level decoder to achieve sub-1.0 RT for real-time video QA applications as part of a four-member research team.
- Collected and curated domain-specific audio/text corpora from target video series to optimize acoustic and language models.
- Conducted acoustic model retraining and language model updates using the new corpora, improving QA accuracy on complex video narratives.

Publications

International Journal

- **J. Oh**, J. Nam, and J.-H. Kim, "HiTCA: Fusing Hierarchical Text and Contextual Audio for Accurate VCR," *Journal on Audio, Speech, and Music Processing*, vol. 2026, article 41, 2026. doi: 10.1186/s13636-026-00481-2. [SCIE]
- S. Ma, **J. Oh**, M. Kim, and J.-H. Kim, "Survey on Deep Learning-based Speech Technologies in Voice Chatbot Systems," *KSII Transactions on Internet & Information Systems* (*TIIS*), vol. 19, no. 5, pp. 1406–1440, 2025. [SCIE]
- **J. Oh**, E. Cho, and J.-H. Kim, "Integration of WFST language model in pre-trained Korean E2E ASR model," *KSII Transactions on Internet and Information Systems* (*TIIS*), vol. 18, no. 6, pp. 1692–1705, 2024. [SCIE]
- S. Seo, **J. Oh**, E. Cho, H. Park, G. Kim, and J.-H. Kim, "TP-MobNet: A Two-pass Mobile Network for Low-complexity Classification of Acoustic Scene," *Computers, Materials & Continua*, vol. 73, no. 2, pp. 3291–3303, 2022. [SCIE]
- M. Lim, D. Lee, H. Park, Y. Kang, **J. Oh**, J.-S. Park, G.-J. Jang, and J.-H. Kim, "Convolutional neural network based audio event classification," *KSII Transactions on Internet and Information Systems* (*TIIS*), vol. 12, no. 6, pp. 2748–2760, Jun. 2018. [SCIE]

International Conference

- S. Ma, **J. Oh**, and J.-H. Kim, "Retention-Preserving Gradient Projection with Entropy-Guided Token-Level Distillation for Rehearsal-Free Continual ASR," in *Proc. Interspeech*, Sydney, Australia, pp. 6599–6603, 2026. doi: 10.21437/Interspeech.2026-2309.
- **J. Oh** and J.-H. Kim, "Adapter-Only Bridging of Frozen Speech Encoder and Frozen LLM for ASR," in *Advances in Knowledge Discovery and Data Mining* (*PAKDD 2026*), Lecture Notes in Computer Science, vol. 16600, pp. 251–262, Springer, Singapore, 2026. [Oral]
- **J. Oh** and J.-H. Kim, "SEAM: Bridging the Temporal-Semantic Granularity Gap for LLM-based Speech Recognition," in *Findings of the Association for Computational Linguistics: EACL 2026*, pp. 2135–2144, 2026.

- **J. Oh**, H. Park, and J.-H. Kim, "Speech Intelligibility Prediction of Dysarthria Using Deep Convolutional Networks," in *Proc. APIC-IST 2023*, pp. 236–237, 2023.
- M. Kim, **J. Oh**, and J.-H. Kim, "Automated Dysarthria Severity Classification Using Diadochokinetic test and Speech Intelligibility Based on LightGBM," in *Proc. APIC-IST 2023*, pp. 12–13, 2023.
- S. Seo, M. Lim, D. Lee, H. Park, **J. Oh**, D. J. Rim, and J.-H. Kim, "Environmental noise robustness for Korean fricatives using speech enhancement generative adversarial networks," in *Proc. IEEE Int. Conf. Big Data and Smart Computing* (*BigComp*), pp. 1–4, 2019.
- S. Seo, D. J. Rim, M. Lim, D. Lee, H. Park, **J. Oh**, C. Kim, and J.-H. Kim, "Shortcut connections based deep speaker embeddings for end-to-end speaker verification system," in *Proc. Interspeech*, pp. 2928–2932, 2019.

Domestic Journal

- 이정필, 장재후, 김지현, 김민섭, 김성준, 김민서, 김하영, **오준석**, 정원, 김장연 외, "음성에 기반한 마비말장애 진단과 설명이 가능한 시스템," 정보과학회지, 42(4): 45–56, 2024. [KSCI/KCI]
- H. Park, Y. Kang, M. Lim, D. Lee, **J. Oh**, and J.-H. Kim, "LFMMI-based acoustic modeling by using external knowledge," *The Journal of the Acoustical Society of Korea*, vol. 38, no. 5, pp. 607–613, 2019. [KSCI/KCI]

Awards

- **2023 한국어 AI 경진대회** (NIA) — Track2-1, 상담 음성인식: 오준석, 김민서, 남주형 (team "상담ONE"), 장려상
- **2022 한국어 인공지능 경진대회** (NIA) — 기업현안(회의음성): 오준석, 김하영 (team "SGCSE"), 최우수상/네이버 대표 (1위)
- **2021 음절인식률 측정 알고리즘 개발 대회** (KT alpha) — 박호성, 오준석, 조은수 (team "검은사케동"), 최우수상 (1위)

Teaching

Audio Recognition, Synthesis & Transformation *Fall 2024*
Teaching Assistant | CSE5109/CSEG109/AIEG109/AIE5109 | Sogang University | Prof. Ji-Hwan Kim
Led lab sessions on audio processing, deep learning basics, language models, and FastSpeech2 TTS using PyTorch and Colab.

Deep Learning-based Automatic Speech Recognition *Summer 2023*
Teaching Assistant | Samsung AI Academy | Sogang University × Samsung Electronics | Prof. Ji-Hwan Kim
Hands-on ASR tutorial covering audio handling, MLP, CTC, Whisper, NVIDIA NeMo fine-tuning, and WFST using PyTorch and Colab notebooks.

Audio Recognition, Synthesis & Transformation *Fall 2023*
Teaching Assistant | CSE5109/CSEG109/AIEG109/AIE5109 | Sogang University | Prof. Ji-Hwan Kim
Led lab sessions on audio processing, PyTorch, RNN/CNN/Seq2Seq, and FastSpeech2/VocGAN TTS using PyTorch and Colab.

Conversational User Interface *Spring/Fall 2022*
Teaching Assistant | CSE5311/CSEG311/GITA370 | Sogang University | Prof. Ji-Hwan Kim
Led lab sessions on dialogue systems and conversational AI interface design.

Patents

- KR 10-2699607 (B1) — *Corpus Construction Service Provision Server and Method* (코퍼스 구축 서비스 제공 서버 및 방법).
  Inventors: Ji-Hwan Kim; Junseok Oh; Hosung Park.
  Assignee: Sogang University Industry–Academic Cooperation Foundation.
  Filed: Apr 29, 2020 (KR 10-2020-0052570); Published: Nov 8, 2021 (KR 10-2021-0133667); Granted: Aug 22, 2024; Gazette: Aug 26, 2024.

Certificates

- NVIDIA Deep Learning Institute Certificate, *Building Conversational AI Applications*, 2022

Technical Skills

Programming Languages & Scripting	Python
Deep Learning Frameworks	PyTorch, NVIDIA NeMo, Hugging Face Transformers, PEFT
E2E ASR Toolkits	Kaldi
Other Tools & Libraries	KenLM